

MIDTERM STUDY GUIDE · CORRECTED EDITION

Calculus Midterm Reviewer

Functions, Limits, Derivatives, Tangents, Polynomial Curves & Time Rates

This edition corrects three verified errors found in the original slide deck (two formula misprints and one unsound inflection-point rule), and adds a worked practice exam with independently audited solutions.

Part 1 · One-Page Cheat Sheet

Corrected version — supersedes the original PDF where they disagree.

1. Functions, Domain & Continuity

- **Domain:** denominator $\neq 0$; even-root radicand ≥ 0 .
- **Piecewise:** pick the branch whose interval condition x satisfies.

Continuous at $x = c$ iff all three hold:

- 1 $f(c)$ is defined
- 2 $\lim_{x \rightarrow c} f(x)$ exists
- 3 $\lim_{x \rightarrow c} f(x) = f(c)$

2. Limits — Order of Attack

- **Direct substitution** — always try first.
- If $0/0$: Factor and cancel · Conjugate/rationalize radicals · L'Hôpital's Rule (differentiate top & bottom, $0/0$ or ∞/∞ only)
- **One-sided:** two-sided limit exists iff left limit = right limit.

3. Derivative Rules

Rule	Formula
Constant	$d/dx (c) = 0$
Power	$d/dx (x^n) = n \cdot x^{n-1}$
Constant multiple	$d/dx (c \cdot x^n) = c \cdot n \cdot x^{n-1}$
Product	$(uv)' = u'v + uv'$
Quotient	$(u/v)' = (u'v - uv') / v^2$
Chain	$d/dx f(g(x)) = f'(g(x)) \cdot g'(x)$

← slide 11 of the original deck prints the “Constant multiple” and “Power” rows without the exponent n on the left-hand side — see Part 2, Errata #1.

Tip: rewrite roots as fractional powers first ($\sqrt{x} = x^{(1/2)}$).

4. Tangent & Normal Lines at (x_0, y_0)

$$m_{\text{tan}} = f'(x_0) \quad \text{Tangent: } y - y_0 = f'(x_0)(x - x_0)$$

$$m_{\text{norm}} = -1/f'(x_0) \quad \text{Normal: } y - y_0 = -1/f'(x_0) \cdot (x - x_0)$$

Edge case: if $f'(x_0) = 0$, tangent is $y = y_0$ and normal is the vertical line $x = x_0$.

5. Curve Analysis

- **Critical numbers:** x where $f'(x) = 0$ or $f'(x)$ undefined.
- **First Derivative Test:** f' goes $+$ $\rightarrow$ $-$ $\Rightarrow$ relative MAX; $-$ $\rightarrow$ $+$ $\Rightarrow$ relative MIN.
- **Second Derivative Test** (at critical point c , $f'(c)=0$): $f''(c) > 0 \Rightarrow$ concave up $\Rightarrow$ local MIN ("holds water"); $f''(c) < 0 \Rightarrow$ concave down $\Rightarrow$ local MAX ("spills water").
- **Inflection point:** where concavity CHANGES SIGN. Solve $f''(x)=0$ for candidates, then test a point on each side. (x^4 at 0 fails this test.)
- **Don't forget:** plug critical x -values back into the ORIGINAL $f(x)$ to get y .

6. Time Rates (Related Rates)

- **Draw & label.** Mark what varies (x , y , V , r) vs. what is constant. Write given rates as dx/dt .
- **Relate.** One primary equation from geometry: $x^2+y^2=z^2$, $V=(1/3)\pi r^2 h$, $A=\pi r^2$, etc.
- **Differentiate wrt t** implicitly, chain rule on every variable: $d/dt[f(x)] = f'(x) \cdot dx/dt$ ($\leftarrow$ slide 11 misprints this); $x^2+y^2=z^2 \Rightarrow 2x(dx/dt) + 2y(dy/dt) = 2z(dz/dt)$ (drops to 0 only when z is constant, e.g. a fixed ladder).
- **Substitute knowns LAST**, then solve.

7. Exam-Day Checklist

- Watch negative signs in the quotient and chain rules.
- Constants differentiate to zero.
- Signs on rates: **+** increasing, **-** decreasing.
- Check your units.

Part 2 · Errata

Every formula, definition, and figure in the original 13-slide PDF was transcribed and independently verified. This section documents what was wrong and what to use instead.

Confirmed Errors (fix before studying)

1. Slide 11 — Power Rule is written wrong [CRITICAL]

Printed: $d/dx (cx) = c \cdot n \cdot x^{n-1}$

The exponent n is missing from the left-hand side. As printed, the LHS is a linear term whose derivative is just c , so the equation is false for every $n \neq 1$.

Correct: $d/dx (c \cdot x^n) = c \cdot n \cdot x^{n-1}$

2. Slide 11 — Time Rates chain rule is written wrong [CRITICAL]

Printed: $d/dt [f(x)] = f'(x) \cdot x/t$

x/t is a plain ratio, not a derivative. The slide's own "Midterm Tip" column correctly says to append dx/dt , so the formula column contradicts the tip.

Correct: $d/dt [f(x)] = f'(x) \cdot dx/dt$

3. Slide 9 — Inflection-point rule drops the sign-change test [MODERATE]

Printed: "Set $f''(x) = 0$ to locate points of inflection."

$f''(x) = 0$ only produces CANDIDATES. Counterexample: $f(x) = x^4$ has $f''(0) = 0$ but no inflection at $x = 0$, because concavity never changes sign. This also contradicts slide 8, which states the rule correctly.

Correct: "Set $f''(x) = 0$ to find CANDIDATE inflection points, then confirm that concavity actually changes sign across each candidate."

Minor Imprecisions (fine in context, worth knowing)

- **Slide 10, step 3:** shows $2x(dx/dt) + 2y(dy/dt) = 0$ for $x^2 + y^2 = z^2$. The " $= 0$ " only holds when z is CONSTANT (e.g. a fixed-length ladder). General form: $2x(dx/dt) + 2y(dy/dt) = 2z(dz/dt)$.
- **Slide 9:** "Set $f'(x) = 0$ to find all potential max/min" ignores critical points where f' is undefined — harmless here because the slide scopes itself to polynomials, which are differentiable everywhere.
- **Slides 6 & 11:** the normal-slope formula $-1/f'(x_0)$ silently assumes $f'(x_0) \neq 0$. If $f'(x_0) = 0$ the tangent is horizontal and the normal is the VERTICAL line $x = x_0$.

Verified Correct (no action needed)

Slides 2–8, 12, 13 — domain rules, 3-part continuity test, limit strategies, constant/power/product/quotient/chain rules, tangent & normal line equations, critical numbers, First &

Second Derivative Tests, concavity, exam tips.

Slide 6's figure was checked against the math: the curve is $y = x^2$ at $(1,1)$, tangent $y = 2x - 1$ (x-intercept 0.5) and normal $y = -0.5x + 1.5$ (y-intercept 1.5, x-intercept 3). The drawing matches exactly.

Part 3 · Practice Exam

17 problems spanning every topic above. Formulas use the corrected versions from Part 2 — not the two typos originally printed on slide 11. Every answer below was independently re-derived by a second reviewer and matched. Try each problem before reading its solution.

Part 1 — Functions, Domain & Continuity

Problem 1 · Domain & Simplification

State the domain of $f(x) = (x + 3) / (x^2 - 9)$, and simplify $f(x)$ fully.

SOLUTION

$x^2 - 9 = 0 \Rightarrow x = \pm 3$, so denominator $\neq 0$ excludes $x = 3$ and $x = -3$. **Domain: all real x except $x = 3$ and $x = -3$.**

$f(x) = (x+3)/[(x-3)(x+3)] = 1/(x-3)$ for $x \neq -3$ (the factor cancels but the domain restriction from the original function still applies — there's a removable hole at $x = -3$, not a valid input).

Problem 2 · Domain of a Radical

State the domain of $g(x) = \sqrt{5 - x}$.

SOLUTION

Radicand must be ≥ 0 : $5 - x \geq 0 \Rightarrow x \leq 5$. **Domain: $(-\infty, 5]$.**

Problem 3 · Piecewise Continuity

Let $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$. Is f continuous at $x = 2$?

SOLUTION

Check the 3-part test at $c = 2$: (1) $f(2) = 3(2) - 1 = 5$ (defined, using the $x \geq 2$ branch). (2) $\lim_{(x \rightarrow 2^-)} f(x) = 2^2 + 1 = 5$; $\lim_{(x \rightarrow 2^+)} f(x) = 3(2) - 1 = 5$ — both sides agree, so $\lim_{(x \rightarrow 2)} f(x) = 5$ exists. (3) $\lim_{(x \rightarrow 2)} f(x) = 5 = f(2)$. ✓

Yes, f is continuous at $x = 2$.

Part 2 — Limits

Problem 4 · Factoring 0/0

Evaluate $\lim_{(x \rightarrow 3)} (x^2 - 9)/(x - 3)$.

SOLUTION

Direct substitution gives 0/0 (indeterminate) $\rightarrow$ factor: $(x^2 - 9)/(x - 3) = (x - 3)(x + 3)/(x - 3) = x + 3$ for $x \neq 3$.
 $\lim_{(x \rightarrow 3)} (x + 3) = 6$.

Problem 5 · Conjugate / Rationalizing

Evaluate $\lim_{x \rightarrow 0} (\sqrt{x+4} - 2)/x$.

SOLUTION

$0/0 \rightarrow$ rationalize (multiply by conjugate $\sqrt{x+4}+2$): $[(\sqrt{x+4}-2)(\sqrt{x+4}+2)] / [x(\sqrt{x+4}+2)] = (x+4-4) / [x(\sqrt{x+4}+2)] = x / [x(\sqrt{x+4}+2)] = 1/(\sqrt{x+4}+2)$.

Limit as $x \rightarrow 0$: $1/(\sqrt{4} + 2) = 1/4$. **Answer: 1/4.**

Problem 6 · L'Hôpital's Rule

Evaluate $\lim_{x \rightarrow \infty} (3x^2 + 5) / (x^2 - 1)$ using L'Hôpital's Rule.

SOLUTION

Form is ∞/∞ . Differentiate top and bottom: $\lim (6x)/(2x) = \lim 3 = 3$. (Same answer as dividing by the highest power of x , which is the usual shortcut — L'Hôpital confirms it.)

Part 3 — Derivative Rules

Problem 7 · Power Rule

Differentiate $f(x) = 5x^4 - 3x^2 + 7$.

SOLUTION

Using $d/dx(c \cdot x^n) = c \cdot n \cdot x^{n-1}$ (the correct power rule — note the deck's slide 11 prints this without the exponent n ; see Part 2, Errata #1) and the constant rule $d/dx(c) = 0$:

$$f'(x) = 5(4)x^3 - 3(2)x + 0 = 20x^3 - 6x.$$

Problem 8 · Product Rule

Differentiate $f(x) = (2x + 1)(x^2 - 3)$ using the Product Rule.

SOLUTION

$$u = 2x+1, u' = 2; v = x^2-3, v' = 2x.$$

$$f'(x) = u'v + uv' = 2(x^2-3) + (2x+1)(2x) = 2x^2-6 + 4x^2+2x = 6x^2 + 2x - 6.$$

Problem 9 · Quotient Rule

Differentiate $f(x) = (3x - 1)/(x + 2)$ using the Quotient Rule.

SOLUTION

$$u = 3x-1, u' = 3; v = x+2, v' = 1.$$

$$f'(x) = (u'v - uv')/v^2 = [3(x+2) - (3x-1)(1)] / (x+2)^2 = [3x+6-3x+1] / (x+2)^2 = 7/(x+2)^2.$$

Problem 10 · Chain Rule

Differentiate $f(x) = (4x^2 + 1)^5$ using the Chain Rule.

SOLUTION

Outer: $u^5 \rightarrow 5u^4$. Inner: $u = 4x^2 + 1 \rightarrow u' = 8x$.

$$f'(x) = 5(4x^2 + 1)^4 \cdot 8x = 40x(4x^2 + 1)^4.$$

Part 4 — Tangent & Normal Lines

Problem 11 · Tangent & Normal at (1,1)

Find the tangent and normal line to $f(x) = x^2$ at the point (1, 1).

SOLUTION

$$f'(x) = 2x \rightarrow f'(1) = 2. \text{ Tangent: } y - 1 = 2(x - 1) \rightarrow y = 2x - 1.$$

$$\text{Normal slope} = -1/2. \text{ Normal: } y - 1 = -1/2(x - 1) \rightarrow y = -0.5x + 1.5.$$

(This matches the worked figure on slide 6 of the reviewer exactly — tangent x-intercept 0.5, normal y-intercept 1.5 and x-intercept 3.)

Problem 12 · Tangent at a Given x

Find the tangent line to $f(x) = x^3 - 2x$ at $x = 2$.

SOLUTION

$$f(2) = 8 - 4 = 4 \rightarrow \text{point } (2, 4). f'(x) = 3x^2 - 2 \rightarrow f'(2) = 12 - 2 = 10.$$

$$\text{Tangent: } y - 4 = 10(x - 2) \rightarrow y = 10x - 16.$$

Part 5 — Curve Analysis (Extrema & Concavity)

Problem 13 · Extrema & Inflection of a Cubic

Find all critical numbers of $f(x) = x^3 - 3x^2 - 9x + 5$, classify each as a relative max or min, and find the point(s) of inflection.

SOLUTION

$f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x-3)(x+1)$. Critical numbers: $x = 3$, $x = -1$ (f' is a polynomial, so no “undefined” cases here).

First Derivative Test: $x < -1$: $f' > 0$; $-1 < x < 3$: $f' < 0$; $x > 3$: $f' > 0$.

f' goes $+$ $\rightarrow$ $-$ at $x = -1 \Rightarrow$ **relative MAX at $x = -1$** . f' goes $-$ $\rightarrow$ $+$ at $x = 3 \Rightarrow$ **relative MIN at $x = 3$** .

y-values: $f(-1) = -1 - 3 + 9 + 5 = 10 \rightarrow$ max point $(-1, 10)$. $f(3) = 27 - 27 - 27 + 5 = -22 \rightarrow$ min point $(3, -22)$.

Inflection: $f''(x) = 6x - 6 = 0 \Rightarrow x = 1$ (candidate). Check sign change: $f''(0) = -6 < 0$ (concave down), $f''(2) = 6 > 0$ (concave up). Sign changes $\Rightarrow$ **genuine inflection point at $x = 1$** , $y = f(1) = 1 - 3 - 9 + 5 = -6$, so **(1, -6)**.

(Per Part 2 Errata #3: always confirm the sign change — don't stop at $f''(x)=0$. Here it does change sign, so $x=1$ is a real inflection point.)

Problem 14 · The x^4 Counterexample

Does $f(x) = x^4$ have an inflection point at $x = 0$?

SOLUTION

$f''(x) = 12x^2 \rightarrow f''(0) = 0$ (candidate). Check sign either side: $f''(-1) = 12 > 0$ and $f''(1) = 12 > 0$ — concave up on BOTH sides, no sign change.

No — $x = 0$ is not an inflection point, even though $f''(0) = 0$. This is exactly the counterexample showing why " $f''(x)=0 \Rightarrow$ inflection" (as misprinted in slide 9's takeaway box) is false without the sign-change check.

Part 6 — Time Rates (Related Rates)

Problem 15 · Sliding Ladder

A ladder 10 ft long rests against a wall. The bottom is pulled away from the wall at 2 ft/s. How fast is the top sliding down when the bottom is 6 ft from the wall?

SOLUTION

1. Draw & label: x = distance from wall to base, y = height of top on wall, $z = 10$ (constant, ladder length). Given $dx/dt = 2$ ft/s. Find dy/dt when $x = 6$.

2. Relate: $x^2 + y^2 = z^2 = 100$.

3. Differentiate wrt t : $2x(dx/dt) + 2y(dy/dt) = 0$ (the right side is 0 here because $z = 10$ is CONSTANT — this is the case where the slide-10 shortcut is actually valid; see Part 2 note).

At $x = 6$: $y^2 = 100 - 36 = 64 \rightarrow y = 8$. $2(6)(2) + 2(8)(dy/dt) = 0 \rightarrow 24 + 16(dy/dt) = 0 \rightarrow dy/dt = -24/16 = -1.5$ ft/s (negative $\Rightarrow$ top is sliding DOWN at 1.5 ft/s).

Problem 16 · Expanding Balloon

A spherical balloon is inflated so that its volume increases at 100 cm³/s. Find the rate of change of the radius when $r = 5$ cm. ($V = (4/3)\pi r^3$)

SOLUTION

1. Given $dV/dt = 100$ cm³/s. Find dr/dt when $r = 5$. **2.** $V = (4/3)\pi r^3$.

3. Differentiate wrt t : $dV/dt = 4\pi r^2 \cdot dr/dt$ (chain rule: $d/dt[f(r)] = f'(r) \cdot dr/dt$ — the correct form; slide 11 misprints this as $f'(x) \cdot x/t$, see Part 2 Errata #2).

$$100 = 4\pi(5^2)(dr/dt) = 100\pi(dr/dt) \rightarrow dr/dt = 100/(100\pi) = 1/\pi \approx 0.318 \text{ cm/s.}$$

Problem 17 · Draining Cone

A cone-shaped tank drains so that $dh/dt = -0.5$ cm/min when $h = 4$ cm and $r = 3$ cm at that instant, with r and h related by $r = (3/4)h$ (a fixed similar-triangle ratio for the tank's shape). Find dV/dt at that instant. ($V = (1/3)\pi r^2 h$)

SOLUTION

1. $r = (3/4)h \Rightarrow$ substitute so V depends only on h : $V = (1/3)\pi[(3/4)h]^2 h = (1/3)\pi(9/16)h^3 = (3\pi/16)h^3$. 2. Already related above.

3. Differentiate wrt t : $dV/dt = (3\pi/16)(3h^2)(dh/dt) = (9\pi/16)h^2(dh/dt)$.

At $h = 4$, $dh/dt = -0.5$: $dV/dt = (9\pi/16)(16)(-0.5) = 9\pi(-0.5) = -4.5\pi \approx -14.1 \text{ cm}^3/\text{min}$ (volume is decreasing, as expected for a draining tank).

Answer Key (quick check)

#	Answer
1	Domain $x \neq \pm 3$; $f(x) = 1/(x-3)$, $x \neq -3$
2	Domain $(-\infty, 5]$
3	Continuous ✓
4	6
5	1/4
6	3
7	$20x^3 - 6x$
8	$6x^2 + 2x - 6$
9	$7/(x+2)^2$
10	$40x(4x^2+1)^4$
11	Tangent $y=2x-1$; Normal $y=-0.5x+1.5$
12	$y = 10x - 16$
13	Max $(-1, 10)$; Min $(3, -22)$; Inflection $(1, -6)$
14	No inflection at $x=0$
15	-1.5 ft/s
16	$1/\pi \approx 0.318 \text{ cm/s}$

17	$-4.5\pi \approx -14.1 \text{ cm}^3/\text{min}$
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